Maintenance Prediction of Bridges using Entity Embedding Neural Networks

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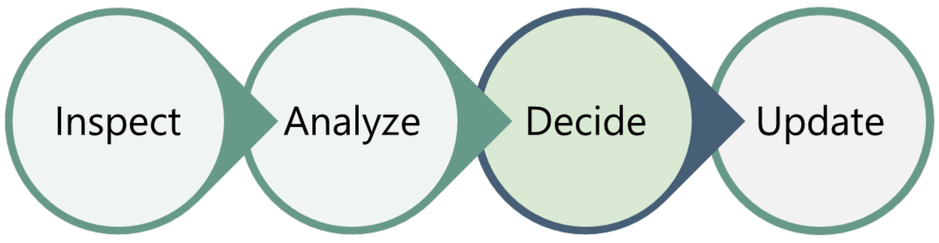
Motivation

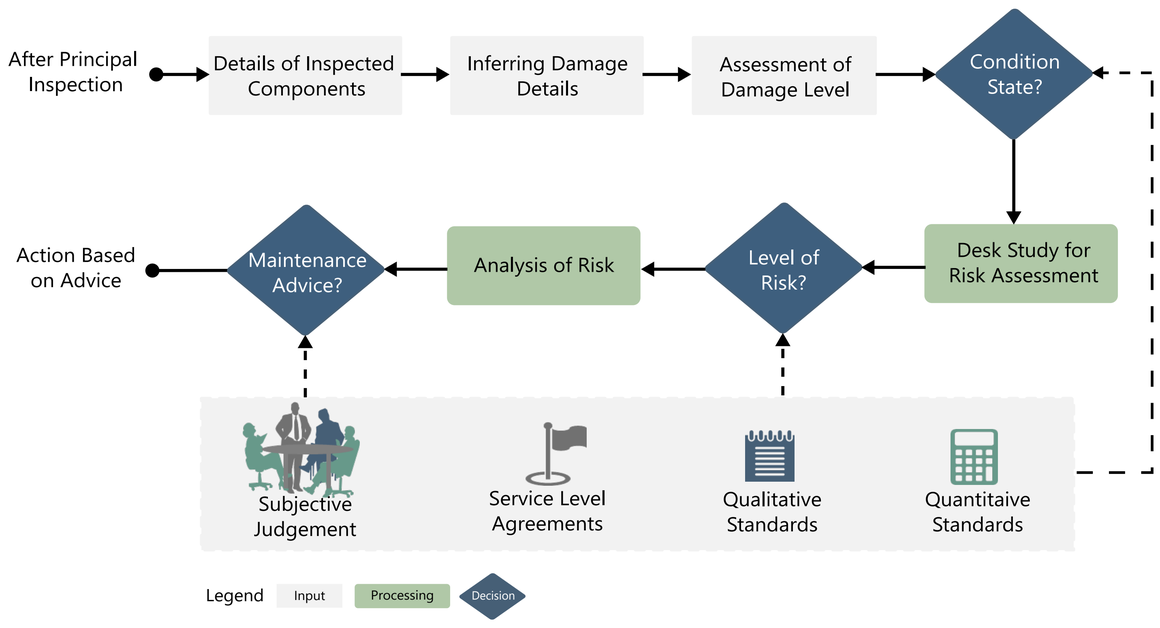
Typical inspection process of civil structures (bridges)



- Subjective assessment by trained inspectors (can vary)
- Difficult to follow, justify and replicate past decisions
- Limited to no access to similar cases of the past
- Little attension towards improving subjective assessment process of inspection

To provide support in subjective assessment of bridges maintenance planning by developing predictive models using historical data

Business process



Inspection to Maintenance Advice process of bridge management system

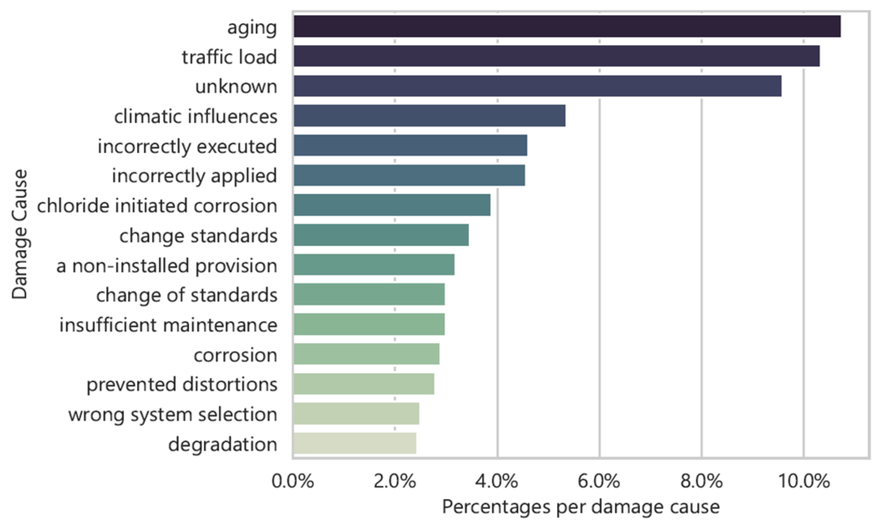
Dataset

Principal inspection data collected from 2007 to 2017

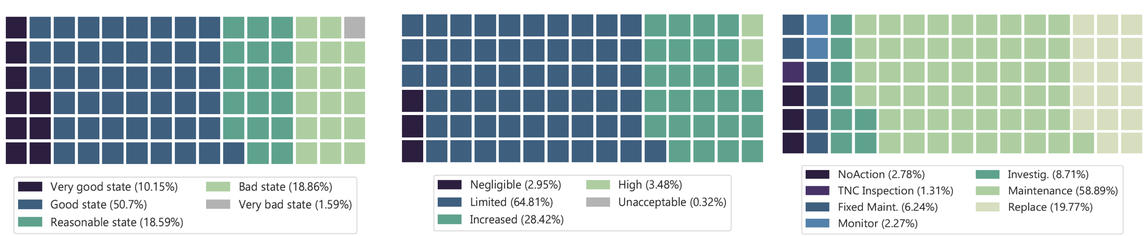
- ✓ Bridges asset register
- ✓ Inspection data
- ✓ Damages detais
- ✓ Risk details

Feature engineering process was guided by experts
Number of features were reduced from 69 to 23 only!

Visual insights

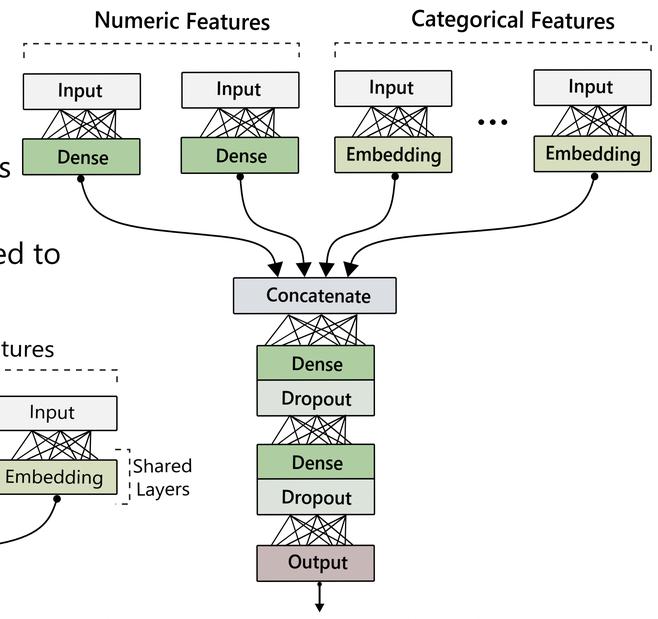


Predictive tasks and data representation

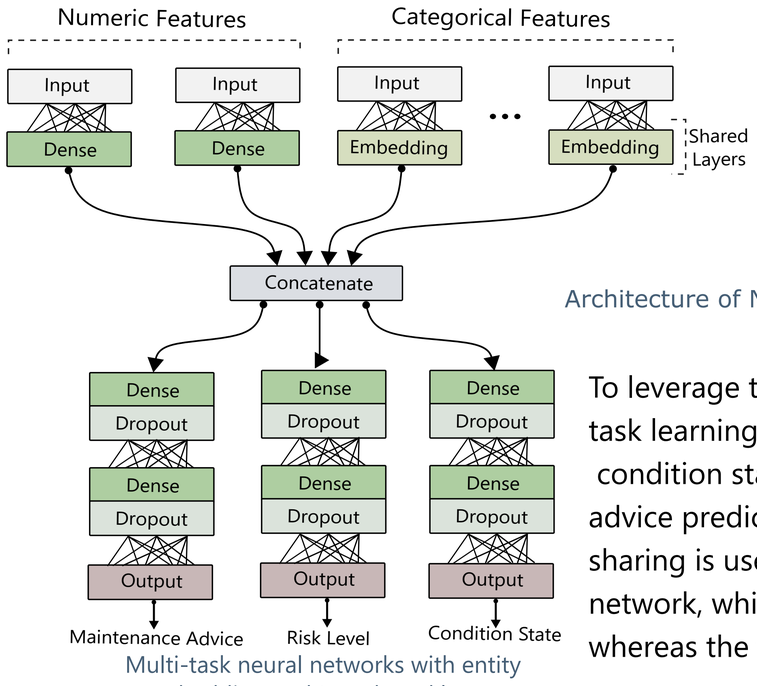


Architecture

A standard three-layers feed-forward neural network was used for all the classification tasks. We utilise entity embedding layer to learn representations from categorical features. Weighted categorical cross-entropy was also applied to handle class imbalance problem.



Architecture of Neural Network with Entity Embeddings



Multi-task neural networks with entity embedding and two shared layers

To leverage task-relatedness, we use multi-task learning to learn unified models for solving condition state, risk level, and maintenance advice prediction tasks. The hard-parameter sharing is used in the initial layers of the network, which are shared across all the tasks, whereas the final layers are problem specific.

Results

Condition state classification with SRS

Classifier	Test set Accuracy	Test set Kappa	Under-sampled majority class Accuracy	Under-sampled majority class Kappa
Gradient Boosted Trees	0.727	0.567	0.732	0.644
NN-EE with class weights NN-EE (cw)	0.796	0.708	0.791	0.724
Multi-task learning – NN (cw)	**0.798**	**0.713**	-	-

Risk level classification with SRS

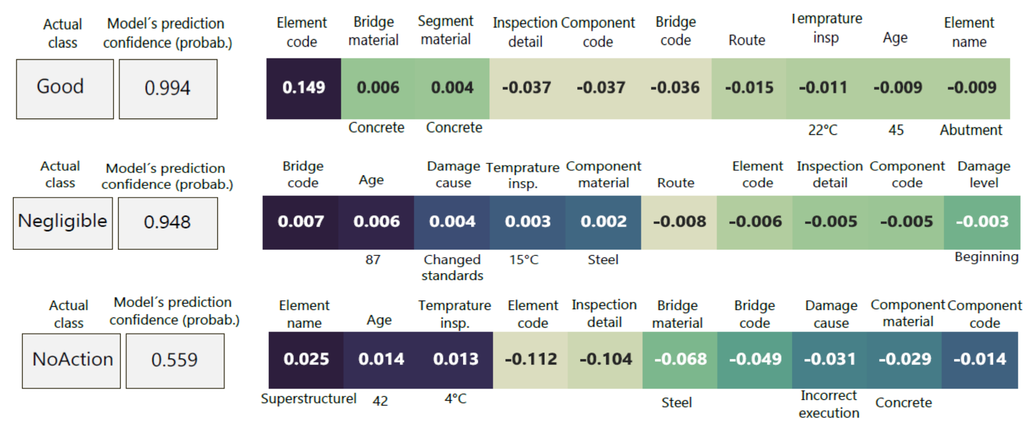
Classifier	Test set Accuracy	Test set Kappa	Under-sampled majority class Accuracy	Under-sampled majority class Kappa
Gradient Boosted Trees	0.807	0.615	0.803	0.664
NN-EE with class weights NN-EE (cw)	0.87	0.762	0.83	0.718
Multi-task learning – NN (cw)	**0.876**	**0.775**	-	-

Maintenance advice classification with SRS

Classifier	Test set Accuracy	Test set Kappa	Under-sampled majority class Accuracy	Under-sampled majority class Kappa
Gradient Boosted Trees	0.809	0.684	0.834	0.775
NN-EE with class weights NN-EE (cw)	0.863	0.79	**0.883**	**0.844**
Multi-task learning – NN (cw)	0.857	0.78	-	-

Interpretability

Explanation of NN-EE (cw) model for all three predictive tasks



Conclusions

- ✓ Aid decision-makers in maintenance related tasks classiifcation with 80% accuracy
- ✓ Utilise data from in-use business process
- ✓ Provide interpretability of the results
- ✓ Multi-task learning for future similar tasks
- ✓ Generic modeling apporach
- ! Evolving data
- ! Data quality
- ! Black-box models

References

Caruana, R. Multitask learning. Machine learning 1997, 28, 41–75.

Guo, C.; Berkhahn, F. Entity embeddings of categorical variables. arXiv preprint arXiv:1604.06737 2016

Ribeiro, M.T.; Singh, S.; Guestrin, C. Why should i trust you?: Explaining the predictions of any classifier. Proceedings of the 22nd ACM SIGKDD 2016

Allah Bukhsh, Z., Stipanovic, I., Saeed, A., & Doree, A. G. Maintenance prediction of bridges using entity embedding neural networks.Under review in Automation in Construction

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